URBAN ANALYTICS AND LABOR ECONOMICS CASE

Buenos Aires Household Survey

This project turns the 2019 Annual Household Survey into a compact case for analytics, consulting, and policy roles. The story is simple: where income differences appear across the city, how much schooling seems to matter, and whether more complex ML models really add that much beyond an interpretable baseline.

5,848
Households aggregated from the survey microdata

13.25%
Estimated labor-income premium per additional year of schooling

0.342
Best out-of-sample ML R-squared on log labor income

What question does it answer?

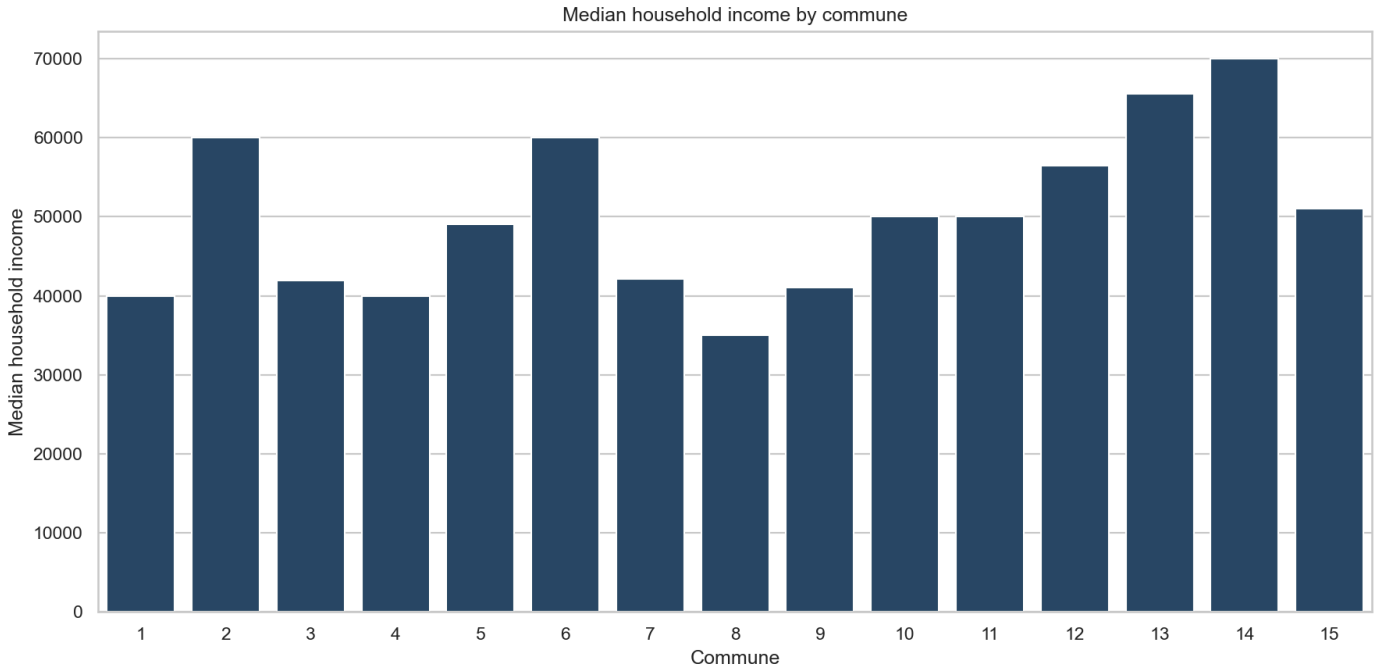
The case asks three practical questions. First, where are the strongest income differences inside Buenos Aires? Second, what happens to labor income once schooling, experience, and gender are read together? Third, if we add ML, do we learn something meaningfully different?

> Executive takeaway: the strongest message is not that some communes are richer. It is that territorial inequality and the labor-income gap can be quantified clearly, while more complex prediction adds only modest extra fit.

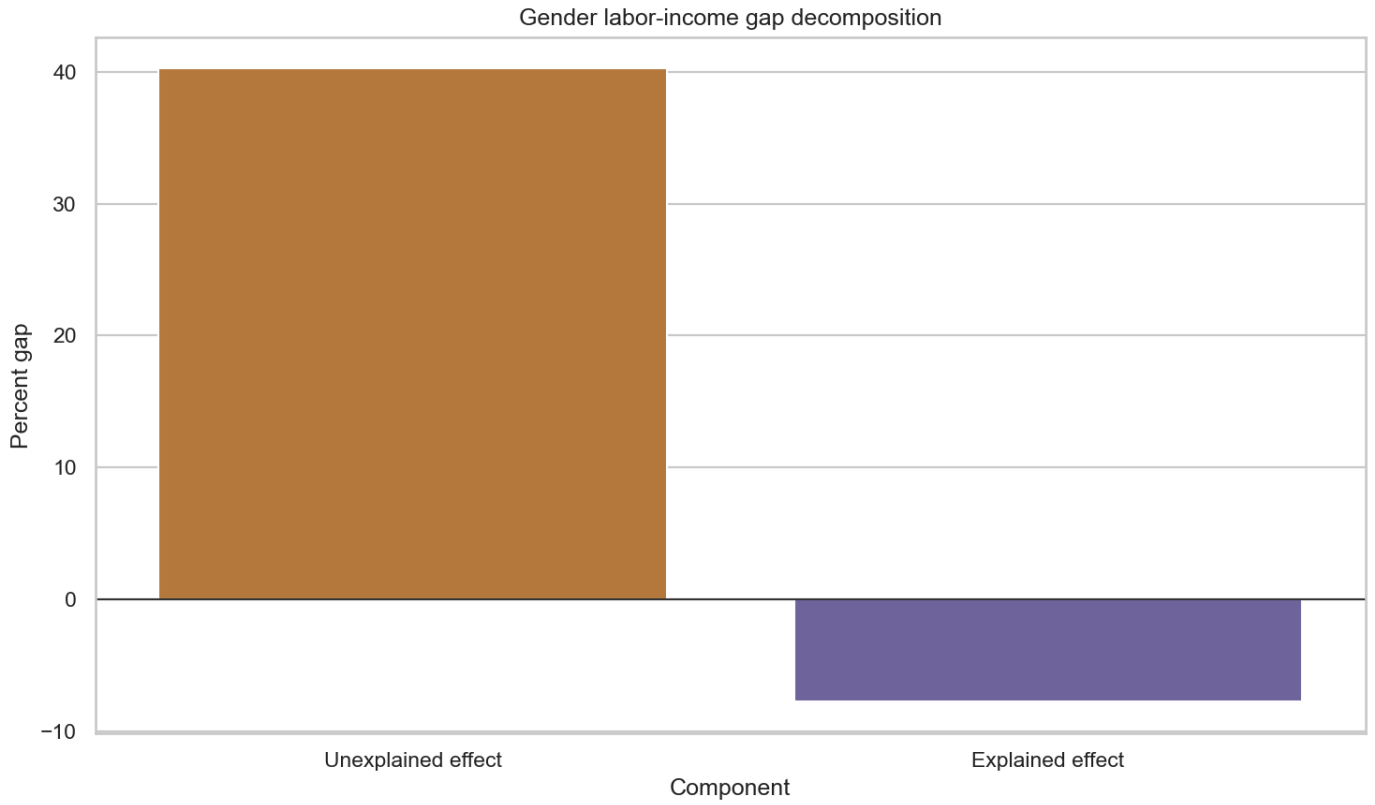
- Commune 14 doubles Commune 8 in median household income: 70,000 versus 35,000.
- Each additional year of schooling is associated with a 13.25% labor-income premium.
- The observed male-female labor-income gap reaches 29.35% in the working sample.

Descriptive and econometric evidence

The first visual shows the city-level income gradient. The second one shows why the gender result matters: once standard observed characteristics are controlled for, the unexplained component of the gap remains large.



Territorial inequality is easy to see. Higher-income communes do not just earn more; they also tend to support smaller households.

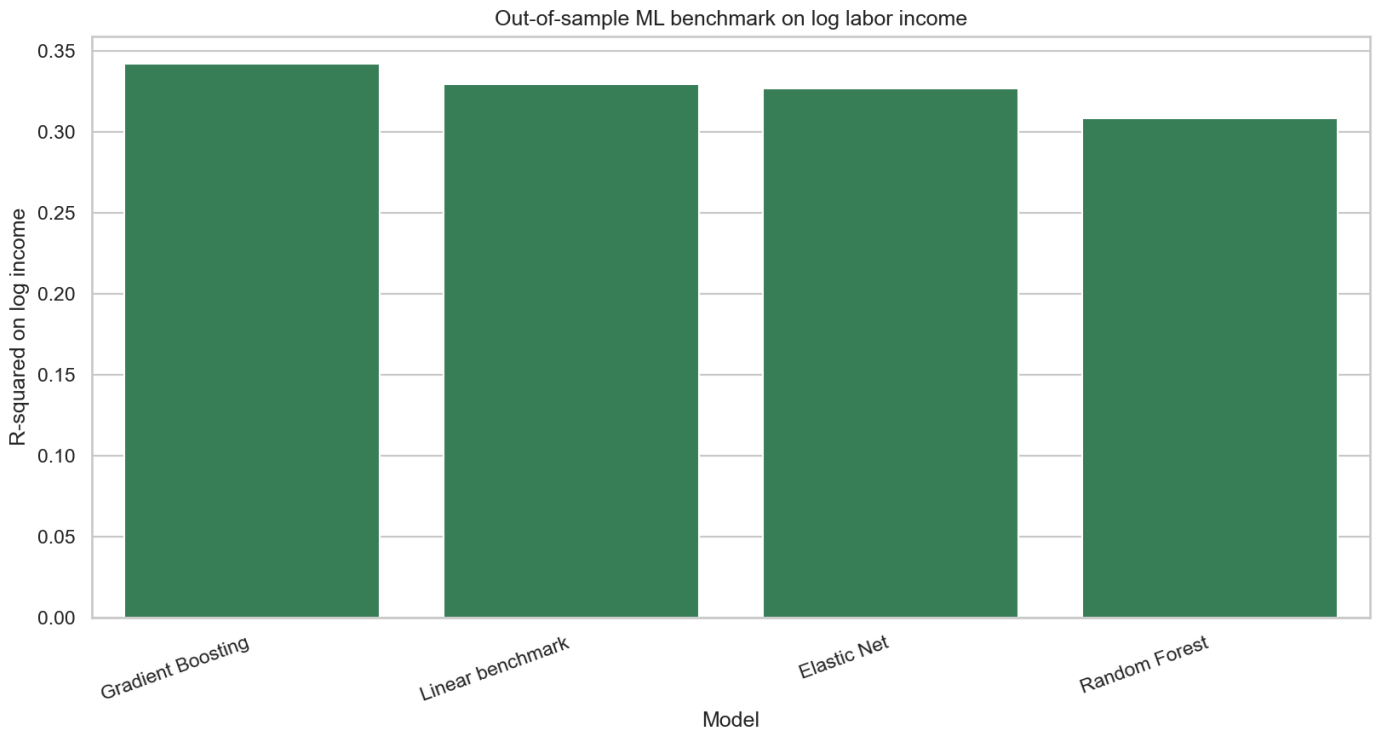


The explained component is negative and the unexplained component is strongly positive. In plain language, the gap is not coming from women in the sample having weaker measured characteristics.

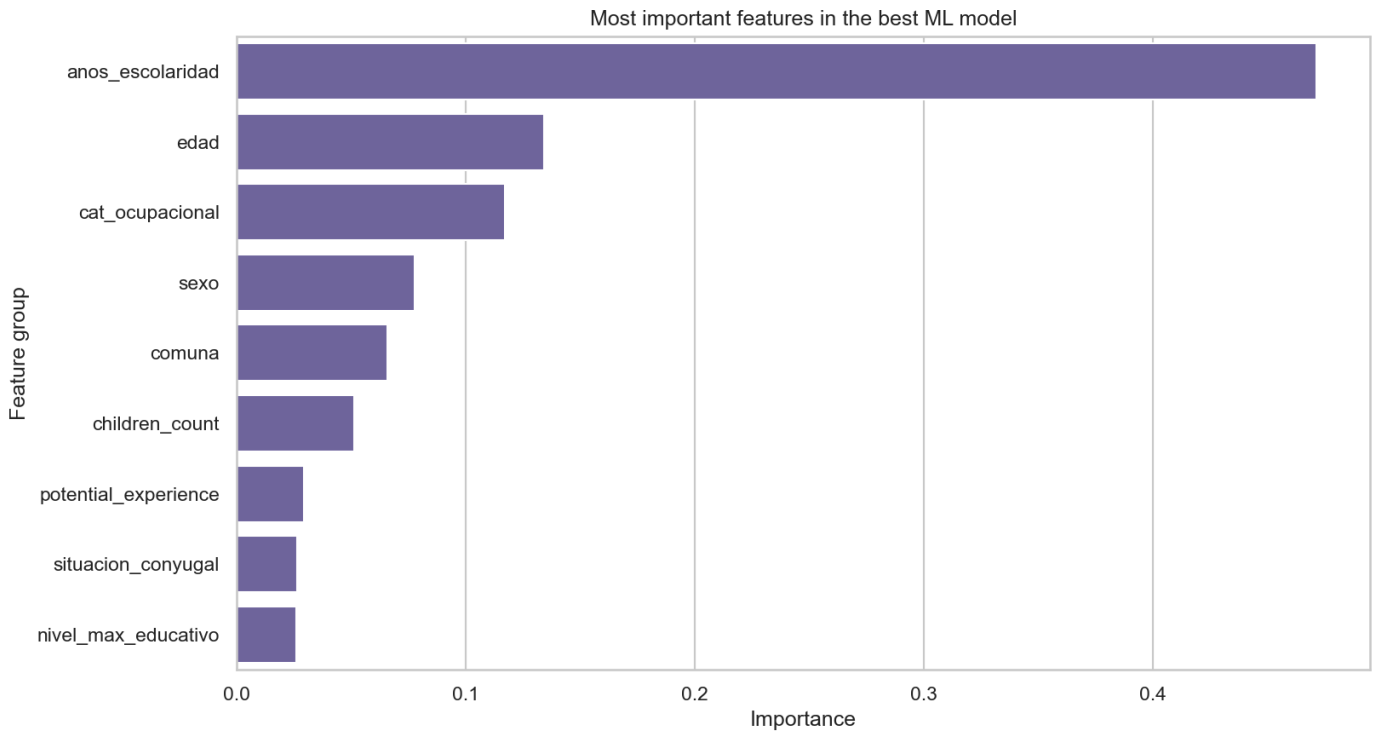
What the ML layer adds

I tested four out-of-sample models on log labor income. The best one is a tuned Gradient Boosting model. It does improve on the linear benchmark, but only slightly. That is useful because it suggests the main signal is already captured by a structured, interpretable specification.

Model	MAE (log income)	RMSE (log income)	R-squared
Gradient Boosting	0.482	0.655	0.342
Linear benchmark	0.488	0.661	0.330
Elastic Net	0.489	0.663	0.327
Random Forest	0.496	0.672	0.309



ML helps a little, but not enough to overturn the main conclusion. The real strength of the project remains interpretability plus economic meaning.



Schooling is still the main feature in the best ML model, followed by age, occupation, gender, and commune. That aligns well with the econometric reading instead of contradicting it.

Why this matters for hiring

This repository works well in interviews because it shows range without becoming noisy. It starts with a real public dataset, moves into an interpretable labor-economics model, and then tests whether more flexible ML meaningfully improves the result.

> Hiring takeaway: this is the kind of project that signals good judgment. It does not just run models. It asks whether extra complexity changes the business or policy story in a meaningful way.